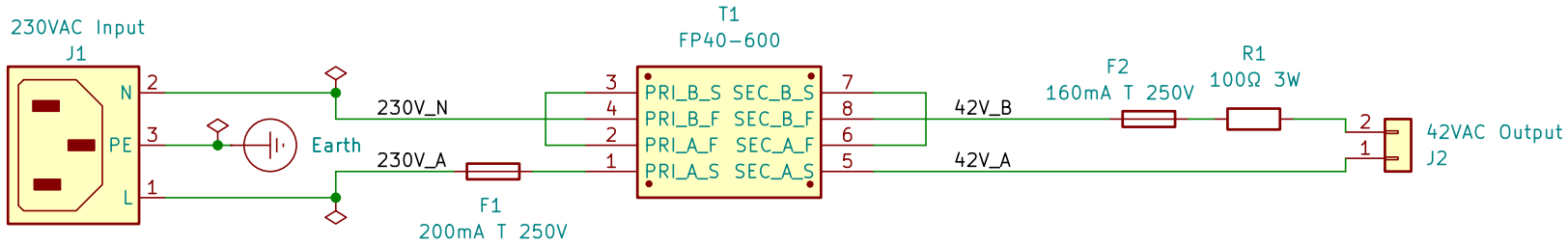
For the Электроника МКШ-2 and 240VAC line voltage, a 100Ω 3W wire wound resistor dropped the output voltage from 50VAC (unloaded) to 42.6VAC.

The design range for the МКШ-2 is 35.7 to 46.2V with nominal 42V.



This design could be repurposed for 115VAC line voltage by modifying the primary inputs, but I can't test that here in Australia.

Refer to the transformer data sheet:
[1] Tie 1 & 3, 2 & 4
[2] Input to 1 & 2

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole
H5 MountingHole
H6 MountingHole

Australian 230VAC Input
IEC 60320 C14 inlet

Soviet 42VAC Output:
Clipsal 402/32 socket or similar (Розетка 42В)

Protective Earth
Bond to metal case

Simple converter for Australian 230VAC mains supply to Soviet 42VAC school supply

Brett Hallen

Sheet: /
File: AU230V-to-SU42V.kicad_sch

Title: 230VAC-to-42VAC Stepdown Transformer

Size: A5	Date: 6/NOV/2026	**Rev: B**
KiCad E.D.A. 10.0.3		Id: 1/1